Graph Neural Network Modeling of Breast MRI Vascular Networks for Treatment Response Prediction

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Introduction + Motivation

Breast cancer is the most common cancer in women worldwide, and for those diagnosed with locally advanced breast cancer (LABC), neoadjuvant chemotherapy (NAC) is the standard of care. However, only 30% of patients achieve a **pathologic complete response (pCR)** to NAC, which is associated with improved survival outcomes. NAC is an intensive treatment that can have significant side effects, and patients who do not achieve pCR may require additional treatment. We would like to develop methods to predict which patients will respond to NAC, so that we can better tailor treatment plans and improve outcomes for patients with LABC.

Ultrafast MRI Dataset

The dataset used in this study consists of **ultrafast breast MRI** scans from $n = 100$ patients with LABC who underwent NAC. The scans were acquired using a 3T MRI scanner and included both pre- and post-contrast images. The dataset also includes clinical information such as age, tumor size, and hormone receptor status, as well as treatment response data (pCR vs. non-pCR).

Graph Representation

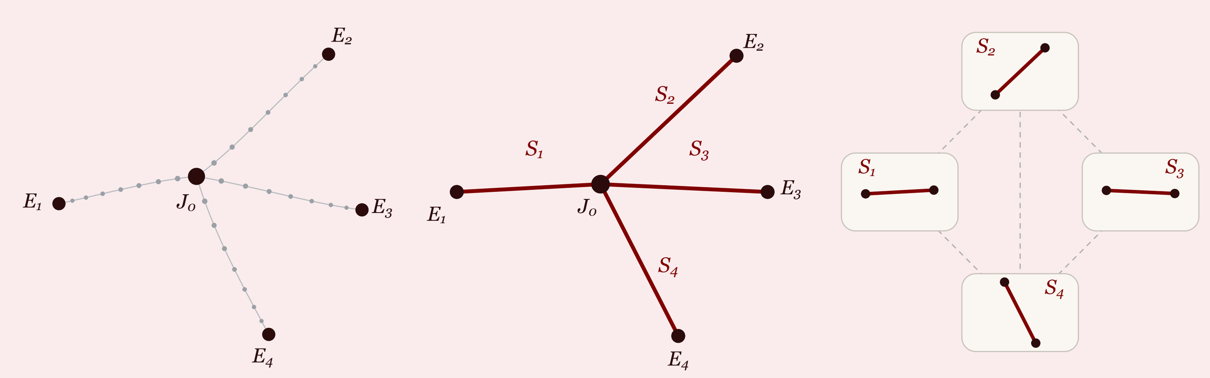

Figure 1. Vascular segments and junctions are converted into graph nodes and edges.

We represent the vascular network of each patient's tumor as a graph, and propose 3 different methods of constructing the graph from the MRI scans. In the following let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph.

- **Voxel-as-node** Let the set of nodes $\mathcal{V}_{\text{vox}}$ be the set of all voxels, and let the set of edges $\mathcal{E}_{\text{vox}}$ be the set of all pairs of adjacent voxels. In this representation, each voxel is a node, and edges connect adjacent voxels.
- **Junction-as-node** Let $\mathcal{V}_{\text{jun}}$ be the set of all junctions (i.e. $e \in \mathcal{E}_{\text{vox}}$ with $\deg(e) > 2$) in the vascular network, and let $\mathcal{E}_{\text{jun}}$ be the set of all pairs of junctions that are connected by a vascular segment.
- **Segement-as-node** Let $\mathcal{V}_{\text{seg}}$ be the set of all vascular segments, and let $\mathcal{E}_{\text{seg}}$ be the set of all pairs of segments that are connected at their endpoints. In this representation, each segment is a node, and edges connect adjacent segments.

Prediction Pipeline and GNN Architecture

Figure 2. End-to-end prediction pipeline.

Vessel segmentation is performed using a **nnU-Net model**, which labels each voxel as vessel or background. The segmentation is simplified into a one-voxel wide network of vessels using the **tc4d** algorithm, then converted into a graph using one of the three representations above.

A **Graph Neural Network (GNN)** predicts treatment response from that graph. Successive **graph convolution (GCN)** layers aggregate information from neighboring nodes, a global pooling layer collapses all nodes into a fixed-size representation of the entire graph, and a fully connected layer maps that representation to a binary prediction (pCR vs. non-pCR).

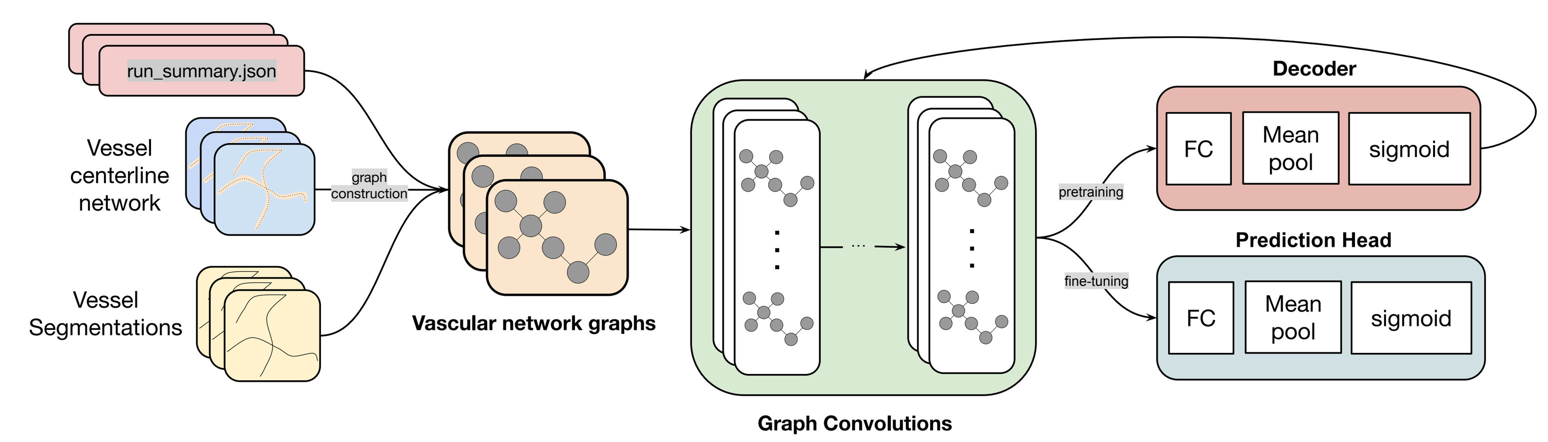

Figure 3. GNN architecture: 4D DCE-MRI through the vessel graph to a pCR probability.

Self-supervised Pretraining

To improve the performance of the GNN, we use a **self-supervised pretraining** approach. The GNN is first trained to predict *per node* the future contrast enhancement, where it learns to predict contrast flow through the network.

Each contrast curve is cut into non-overlapping windows. Shown the frames on an input horizon $[t_1, t_L]$, the model predicts every node's enhancement over the following horizon $(t_L, t_{L+H}]$, scored by mean absolute error against the measured signal. Predictions are conditioned on **elapsed seconds** $\tau = t - t_1$ rather than on frame index, so the irregular ultrafast cadence is handled directly.

Results

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Reproducibility

In an effort to promote reproducibility and transparency in our research, we have made our code publicly available on GitHub. The repository includes the code for the vessel segmentation, graph construction,' GNN training, and evaluation, as well as instructions for how to reproduce our results.

Conclusion

In this study, we proposed a novel approach for predicting treatment response in patients with LABC using ultrafast breast MRI scans and GNNs. Our method demonstrates the ability to predict pCR from the vascular network of the tumor, and that self-supervised pretraining can improve the performance of the GNN. Our results suggest that GNNs can be a powerful tool for predicting treatment response in breast cancer, and may have important implications for personalized treatment planning.

References

[1] Siyi Tang, Jared A. Dunnmon, Khaled Saab, Xuan Zhang, Qianying Huang, Florian Dubost, Daniel L. Rubin, and Christopher Lee-Messer. Self-supervised graph neural networks for improved electroencephalographic seizure analysis, 2022. URL `https://arxiv.org/abs/2104.08336`.